



Exploring the influence of competition and collaboration on learning performance in digital game-based learning

Ding-Chau Wang¹ · Yong-Ming Huang² 

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Abstract

Competition in digital game-based learning (DGBL) is tantamount to a double-edged sword, because it not only improves students' learning performance, but may also cause them psychological stress. Therefore, researchers focus on collaboration, for it can mitigate the negative effect of competition. However, studies on the correlations among competition, collaboration, and learning performance in DGBL remain wanting so far. To remedy this deficiency, this study developed a question bank practice game which offers a situation of intergroup competition with intragroup collaboration, and devised a research model to investigate the correlations among competition, collaboration, and learning performance in DGBL. The research findings of this study indicated that: (1) perceived competition is a factor more significant than perceived collaboration behind learning performance, in which perceived competition directly affects perceived collaboration; (2) perceived competition not only directly affects behavioral intention, but also indirectly influences it via perceived enjoyment, whereas perceived collaboration simply has indirect influence on behavioral intention through perceived usefulness; and (3) behavioral intention directly influences learning performance. These research findings showed that the combination of competition and collaboration indeed influences students' learning performance in DGBL, and competition occupies the most crucial role. This is because the pleasure brought by competition strongly prompts students to engage in DGBL, which in turn influences their learning performance in DGBL. Meanwhile, competition also facilitates students' collaboration, from which they benefit personally in terms of learning.

Keywords Competition · Collaboration · Learning performance · Digital game-based learning

✉ Yong-Ming Huang
ym.huang.tw@gmail.com

Ding-Chau Wang
dcwang@stust.edu.tw

¹ Department of Information Management, Southern Taiwan University of Science and Technology, Tainan City, Taiwan (ROC)

² Department of Multimedia and Entertainment Science, Southern Taiwan University of Science and Technology, Tainan City, Taiwan (ROC)

Introduction

Competition has been widely used as a promising way to boost students' learning performance in digital game-based learning (DGBL) (Chen et al., 2020; Dindar et al., 2021; Yang et al., 2020). DGBL is a learning process that integrates learning contents into digital games, which enables students to acquire knowledge through game-playing (Chen et al., 2020; Yang et al., 2020). In DGBL, competition has been extensively employed to encourage students' engagement in learning. Competition is an act of comparing one's own performance with others' on the same task at the same time (Chen & Chiu, 2016; Hwang & Chang, 2016). In other words, competition is a form of social comparison, through which people assess their self-worth (Chen & Chang, 2020; Nebel et al., 2016). For social comparison, people are willing to invest effort in increasing their capability so as to ensure their own superiority. Once they outperformed their rivals, a sense of achievement ensues, and they will recognize their self-worth (Hwang et al., 2019; Nebel et al., 2016). Accordingly, competition is deemed an effective impetus in DGBL (Chen & Chang, 2020; Chen et al., 2020; Dindar et al., 2021). For example, a leaderboard is a common competition mechanism. It allows students to discern their own and their rivals' performance in real-time. When students noticed that they outperformed their peers, they will feel excited and gain a sense of achievement, which will in turn facilitate students' continuous engagement in learning (Dindar et al., 2021; Hwang et al., 2019; Nebel et al., 2016). So far, competition in DGBL has been widely applied to improving students' learning motivation, engagement, and effectiveness (Burguillo, 2010; Cagiltay et al., 2015; Chen & Chang, 2020).

On the other hand, collaboration in DGBL has gradually come under scholarly spotlight, for it can mitigate the negative effect of competition (Chen, 2019; Dindar et al., 2021; Hung et al., 2015; ter Vrugte et al., 2015). In fact, related studies have shown that competition cannot always galvanize students in learning in an effective way, since students may concentrate on winning the game without the intention to grasp the knowledge embedded in it (Chen et al., 2019; Riemer & Schrader, 2022). Besides, competition may even undermine students' mental health, because they may directly discern that they are inferior to others in competition, hence frustration, tense, and anxiety (Dindar et al., 2021; Yang et al., 2020). Therefore, researchers have tried to use collaboration to palliate the negative effect of competition (Chen, 2019; Dindar et al., 2021; Hung et al., 2015; ter Vrugte et al., 2015). Collaboration is an act of working together to reach a shared goal (Dindar et al., 2021; Hung et al., 2015). To put it another way, collaboration is tantamount to social support or mutual concern that helps people accomplish a common objective (Huang, 2020a; Daradoumis et al., 2006). Collaboration in DGBL has been recognized to be able to buffer the negative effect resulting from competition, because the social support among students can alleviate their stress and apprehension when confronting challenges (Dindar et al., 2021; Liu & Hung, 2016). More important, collaboration can improve students' learning performance (Hung et al., 2015; ter Vrugte et al., 2015). In DGBL, collaboration enables students to acquire knowledge collaboratively through discussion, negotiation and idea exchange, thereby solving the challenges in the game. Meanwhile, students' interaction will transform their learning approach from memorizing into creation, through which their cognitive structures will produce new ideas that help them internalize the knowledge embedded in the game (Chen, 2019; Hung et al., 2015; ter Vrugte et al., 2015).

However, very few studies delved deep into the correlations among competition, collaboration, and learning performance in DGBL. It is an issue worth investigation, because competition and collaboration have been frequently employed in DGBL for the purpose

of improving students' learning performance. That is, DGBL will be applied and developed in a more effective fashion if we can clarify this issue. So far, the majority of literature focused simply on the correlation between competition and learning performance, and those studies differed partly in results (Cagiltay et al., 2015; Chen et al., 2019). For example, Cagiltay et al. (2015) found that competition has an influence on students' engagement in DGBL, and students' engagement in DGBL indirectly affects their learning performance, whereas Chen et al. (2019) showed that competition does not affect students' engagement in DGBL, but students' engagement in DGBL has an influence on their learning performance. These results suggested that further investigation is necessary if we are going to clarify the correlation between competition and learning performance. Of all the elements in DGBL, none are so distinctive as competition, because it may be detrimental to learning (Dindar et al., 2021; Yang et al., 2020) in addition to its incentive function (Burguillo, 2010; Cagiltay et al., 2015; Chen & Chang, 2020). In view of the negative effect of competition, researchers refocused their attention onto the role of collaboration in DGBL. For example, Chen and Chiu (2016) found that intergroup competition with intragroup collaboration is effective in enhancing students' engagement and their learning performance, and Chen (2019) also showed that intergroup competition with intragroup collaboration is useful to promote both students' interest and performance in learning. Nonetheless, these studies did not explain how competition and collaboration affect learning performance in an interactive way, even though they have demonstrated that competition with collaboration indeed has an influence on learning performance. Accordingly, the inquiry into the correlations among competition, collaboration, and learning performance has been an imperative for filling the gap in the existing literature.

We developed a question bank practice game which offers a situation of intergroup competition with intragroup collaboration (IC2), and devised a research model to investigate the influences of competition and collaboration on learning performance in DGBL. This IC2 game immersed the participants in game-based question bank practice. In fact, item bank systems have been proved helpful for students' learning, because they make students aware of their learning progress immediately, such as examining which part they have mastered and which part they have not (Hwang et al., 2019). However, students may easily find it tedious and vapid if they just repetitively use this system for learning. As a result, researchers incorporated game elements into this kind of system to overcome this problem (Hwang et al., 2019). Apart from allowing the participants to compete in answering the questions, the IC2 game developed in this study also required them to collaborate in solving the challenges in the game. Thus, the IC2 game not only brought the participants delight in competition, but also mitigated its negative effect. Then we devised a research model by reference to the attributes of the IC2 game. In this model, competition and collaboration were treated as independent variables, while learning performance as the dependent one. Based on this research model, we designed a questionnaire to collect the participants' opinions about the IC2 game; and, according to the data collected from the questionnaire, we analyzed how competition and collaboration affect learning performance in DGBL. In sum, this study contributed to the theory-building and practice in DGBL. In terms of theory-building, previous studies (Chen & Chiu, 2016; Chen, 2019) merely indicated that intergroup competition with intragroup collaboration can influence learning performance in DGBL. We advanced this finding by offering new insights into the key process as to how IC2 affects learning performance in DGBL. In terms of practice, we proposed several concrete suggestions according to the results of our analysis, insofar as to support researchers to explore DGBL in greater detail, assist practitioners in carrying out DGBL more effectively, and help developers create better digital games.

Theoretical background and related studies

In recent years, the key factors behind learning performance have come under the scholarly spotlight for the purpose of advancing educational technology (Al-Rahmi et al., 2018; Garcia et al., 2019; Hwang & Kuo, 2015; Lin et al., 2021). Educational technology is intended to facilitate students' learning and improve their learning performance, and many researchers have proved its usefulness in this regard (Burguillo, 2010; Cagiltay et al., 2015; Chen & Chang, 2020). Nevertheless, using educational technology to boost learning performance is little less than a complex action of technology utilization. Specifically speaking, students' intention to use educational technology is one of the prerequisites for the technology to improve their leaning performance (Huang, 2017, 2019, 2020a). That is, when students have no intention to use educational technology, it will be ineffective no matter how powerful its function is. Accordingly, many researchers investigated the crucial factors behind learning performance from the angle of the technology acceptance model (TAM), in order to foster the development of educational technology (Al-Rahmi et al., 2018; Garcia et al., 2019; Hwang & Kuo, 2015; Lin et al., 2021). Specifically speaking, the TAM originated from the theory of reasoned action (TRA) (Fishbein & Azjen, 1975). Early on, Fishbein and Azjen (1975) proposed the TRA, arguing that attitude and subjective norm directly affect behavioral intention. Attitude refers to "a person's general feeling of favorableness or unfavorableness toward a given stimulus object" (Fishbein & Azjen, 1975: 216). Subjective norm refers to "the person's perception that most people who are important to him think he should or should not perform the behavior in question" (Fishbein & Azjen, 1975: 302). Behavioral intention refers to "a person's subjective probability that he will perform some behavior" (Fishbein & Azjen, 1975: 288). Later, based on the TRA, Davis and his colleagues (Davis, 1989; Davis et al., 1989) added perceived ease of use and perceived usefulness to the TRA, removed subjective norm from it, and finally named this model as the TAM. Accordingly, the TAM comprises four constructs, including perceived ease of use, perceived usefulness, attitude toward using, and behavioral intention. Perceived ease of use refers to the degree to which a person believes that using a particular technology would be free from effort (Davis, 1989). Perceived usefulness refers to the degree to which a person believes that using a particular technology would enhance his or her job performance (Davis, 1989). Attitude toward using refers to a person's general feeling of favorableness or unfavorableness toward using a particular technology (Fishbein & Azjen, 1975). Behavioral intention refers to a person's subjective probability that he will use a particular technology (Fishbein & Azjen, 1975). Davis et al. argued that when a technology is easy to use and contributive to work performance, it will make a favorable impression on users and prompt them to use it, and users may very likely use it; to wit, accept it. Following Davis' finding, educationalists claimed that students' learning performance will be improved if they intend to use educational technology, because they can benefit substantially from using it. For example, Hwang and Kuo (2015) found that students' intention to use a given web-based learning system significantly affects their learning performance. Besides, Al-Rahmi et al. (2018) showed that both the ease of use and the usefulness of social media significantly influence students' intention to use social media, which in turn affects their learning performance. Furthermore, Garcia et al. (2019) indicated that the usefulness of blogs indirectly affects the learning performance of students who use them. Recently, Lin et al. (2021) proved that both the ease of use and the usefulness of augmented reality (AR) indirectly influence students' learning performance via the mediation of students' intention to use AR.

The use of educational games is different from the use of general educational technology, and perceived enjoyment and perceived usefulness are the archetypal constructs adopted to investigate the action of using educational games. On a more specific basis, educational games simultaneously facilitate learning and provide entertainment, which is why their use and acceptance differ from those of general educational technology. When it comes to the use of educational games, researchers tend to consider both of these two aspects. Perceived enjoyment refers to the degree to which a person believes that the experience in using a technology is enjoyable (Davis et al., 1992), which serves as an intrinsic motivation. It implies that users perform an action because they perceive that the process of performing the action is inherently entertaining (Davis et al., 1992). Perceived usefulness serves as an extrinsic motivation. It means that users perform an action because they perceive that performing the action helps achieve a desired outcome (Davis et al., 1992). Previous studies have pointed out that perceived enjoyment is the key factor for users to utilize hedonic technology, and perceived usefulness the major factor behind the use of utilitarian one (Van der Heijden, 2004). Different from other educational technology, educational games perform not only hedonic function (i.e., playing game) but also utilitarian one (i.e., gaining knowledge). It implies that perceived enjoyment and perceived usefulness play the pivotal roles in affecting students' intention toward educational games when educational games are treated as learning tools. For example, Huang (2018) found that perceived enjoyment and perceived usefulness have direct and indirect influences on students' intention to use educational games. Besides, Huang (2019, 2020b) showed that perceived enjoyment plays either a direct or an indirect role in affecting students' intention to use educational games. Recently, Zulfiqar et al. (2021) further proved that both the enjoyment and the usefulness of educational games indirectly influence students' learning performance via the mediation of students' intention to use educational games.

The great effort devoted notwithstanding, there is a surprising paucity of studies on how competition and collaboration influence learning performance in DGBL contexts. Specifically speaking, DGBL not only helps students acquire knowledge, but also brings them pleasure. Competition is a common and effective way to derive pleasure, because students may feel the sense of excitement and achievement by outperforming others in competition (Burguillo, 2010; Hwang et al., 2019; Liu et al., 2013). However, if educational games put undue stress on competition, they may urge students to concentrate on winning the game and ignored the essence of DGBL, namely, learning. This is why researchers gradually set great store by collaboration and apply it to DGBL. In terms of learning performance, the question as to how competition and collaboration influence students' performance in DGBL contexts remains unanswered. To remedy this deficiency, this study established a research model by reference to the literature review in this section. It treats competition and collaboration as the independent variables while learning performance as the dependent one, so as to examine their correlations.

Research design

Research model and hypotheses

Figure 1 is the research model constructed by reference to the literature review in "Theoretical background and related studies" Section. Perceived competition refers to the extent to which students subjectively perceive themselves competing with other

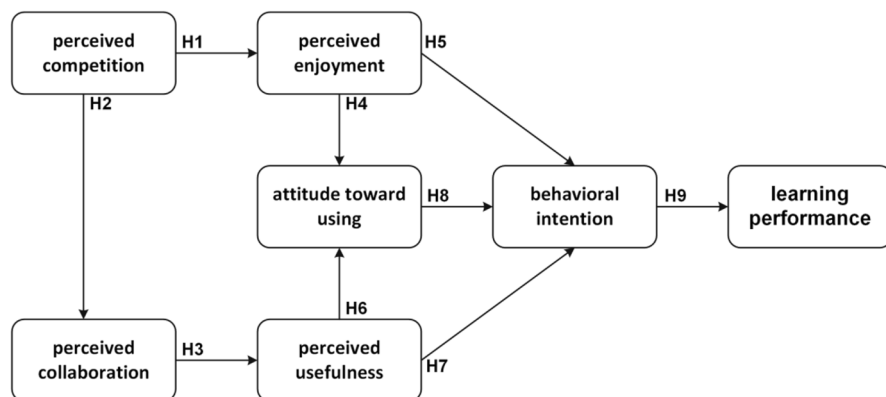


Fig. 1 Research model

teams (Mitchell, 1996). Perceived collaboration refers to the extent to which students subjectively perceive themselves collaborating with their teammates (Tjosvold, 1988). Perceived enjoyment is the extent to which students subjectively enjoy using the IC2 game (Huang, 2018, 2019, 2020b). Perceived usefulness refers to the extent to which students subjectively feel themselves benefiting from using the IC2 game (Huang, 2018, 2019, 2020b). Attitude toward using refers to students' positive or negative feelings about using the IC2 game (Huang, 2018, 2019, 2020b). Behavioral intention refers to the extent to which students subjectively want to use the IC2 game (Huang, 2018, 2019, 2020b). Learning performance refers to the extent to which students subjectively perceive that their learning performance has improved as a result of using the IC2 game (Liao et al., 2015; Xu et al., 2014). Overall, treating perceived competition and perceived collaboration as potential factors behind learning performance in DGBL contexts, we expect this model to contribute to the growing body of literature on DGBL. This model comprises a total of nine hypotheses to be explicated below.

Researchers have not only pointed out that perceived competition and perceived collaboration may significantly influence perceived enjoyment and perceived usefulness, but also extensively discussed the correlations among these factors. Specifically speaking, the existing literature has proved that competition in DGBL directly brings students a sense of pleasure, because it immerses them deeply in peer contest. Students will gain a sense of joy and self-assurance once they capture victory over their rivals (Chen & Chang, 2020; Chen et al., 2020; Dindar et al., 2021). Researchers also noticed that competition can stimulate intragroup collaboration in an IC2 situation, because members' shared goal increases their interdependence and forges their consensus (Cagiltay et al., 2015; Chen & Chiu, 2016; ter Vrugte et al., 2015). Moreover, collaboration has been recognized as useful in facilitating students' learning within DGBL contexts, because it allows students to engage in collaborative learning, which in turn prompts them to notice the usefulness of DGBL (Chen, 2019; Hung et al., 2015; ter Vrugte et al., 2015). Accordingly, the first three hypotheses are formulated as follows:

H1 Perceived competition has a positive effect on perceived enjoyment.

H2 Perceived competition has a positive effect on perceived collaboration.

H3 Perceived collaboration has a positive effect on perceived usefulness.

Researchers have not only identified perceived enjoyment and perceived usefulness as influential on attitude toward using and behavioral intention in DGBL contexts, but also systematically explored their correlations. Specifically speaking, previous studies have indicated that the sense of pleasure students derived from playing a given educational game will positively affect their emotions, and they will express a favorable attitude toward the game as well as intend to use it (Huang, 2018, 2019). Students will also do so when they find the game helpful for their learning (Liang & Yeh, 2008; Zulfiqar et al., 2021). Accordingly, the fourth to seventh hypotheses are formulated as follows:

H4 Perceived enjoyment has a positive effect on attitude toward using.

H5 Perceived enjoyment has a positive effect on behavioral intention.

H6 Perceived usefulness has a positive effect on attitude toward using.

H7 Perceived usefulness has a positive effect on behavioral intention.

Many researchers have investigated and confirmed the correlations among attitude toward using, behavioral intention, and learning performance. To be more specific, scholars have found that students will accept and use a given educational technology when they have a positive attitude toward using it (Huang, 2016, 2018; Liang & Yeh, 2008). The educational technology can ergo improve students' learning performance once they accept and use it (Hwang & Kuo, 2015; Lin et al., 2021; Zulfiqar et al., 2021). In this study, no sooner did the participants cultivate a positive attitude toward the IC2 game we designed, than they decided to learn through using it. As they began to use the game for learning, they immersed themselves in the digital game-based question bank practice without growing weary of it. Therefore, the last two hypotheses are formulated as follows:

H8 Attitude toward using has a positive effect on behavioral intention.

H9 Behavioral intention has a positive effect on learning performance.

The IC2 game

The IC2 game we designed is a question bank practice game. Based on Tetris, it incorporates the functions of question bank practice and the mechanism of intergroup competition. Specifically speaking, the IC2 game is a 2v2 competition game. Each team plays two roles, including a respondent and a gamer. The respondent is responsible for answering the questions randomly put by the question bank. The gamer is responsible for accomplishing the tasks in the game, that is, completing lines by moving differently shaped tetrominoes that descend onto the playing field. During the game, players in the same team can swap their roles and collaborate freely, including mutual consultation, so as to produce optimal performance of the team. Within this context, players can not only fulfill intragroup collaboration, but also experience intergroup competition.

Figures 2 and 3 graphically illustrate the IC2 game. Figure 1a shows the screen of the respondent. The randomly given question and the corresponding options appeared on the

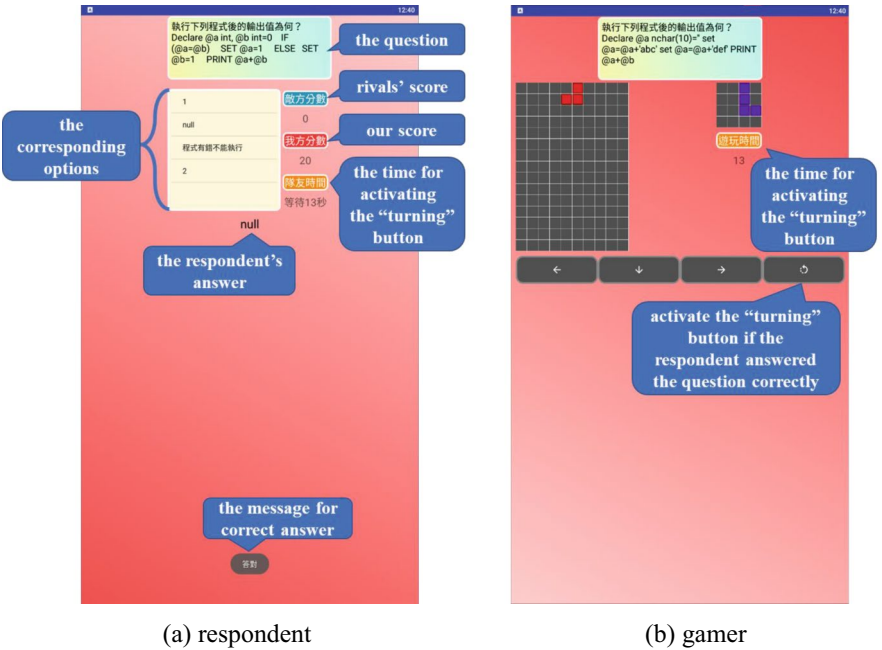


Fig. 2 The screens of submitting the correct answer

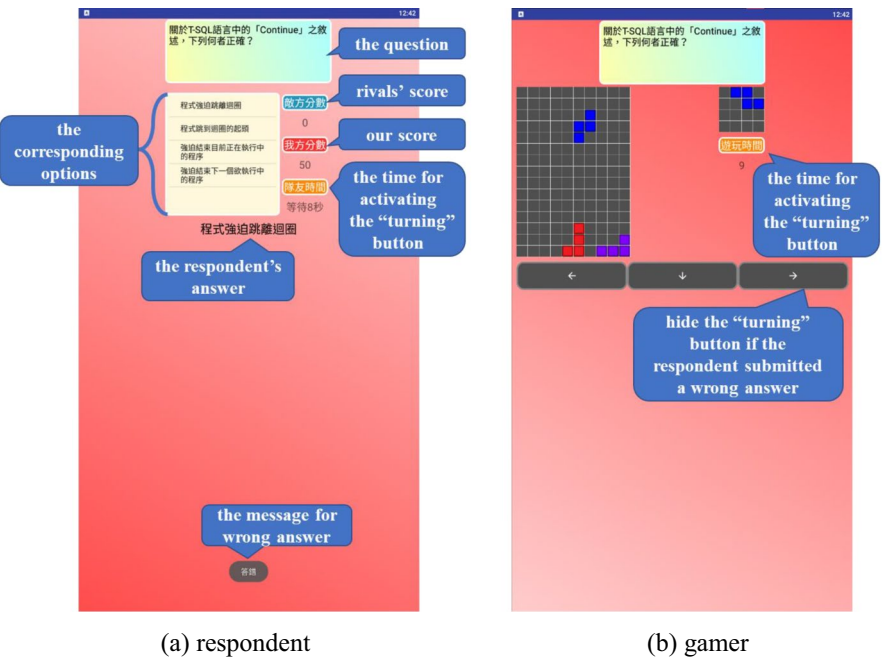


Fig. 3 The screens of submitting a wrong answer

upper part of the screen. The respondent had to choose the correct answer from these options within 70 s. The group earned a score of 20 if the respondent answered the question correctly, which gave the gamer 15 s to activate the “turning” button and complete lines of tetrominoes, as shown in Fig. 1b. Besides, within the 15 s, the group got additional 5 points for each completed line. Contrarily, the group got no score if the respondent submitted a wrong answer, which gave the gamer merely 10 s to complete lines of tetrominoes without the right to activate the “turning” button (shown in Fig. 3). Furthermore, the game required the two participants to swap their roles every 10 questions, making sure that each participant experienced all the types of tasks. Even though the respondent and the gamer were assigned different tasks, they were allowed to discuss with each other over the challenges they respectively confronted during game-playing in order to optimize their respective performance. In other words, this game enabled the participants to simultaneously experience the excitement of intergroup competition and the teamwork of intragroup collaboration, thereby immersing them in game-based question bank practice.

Measurement

Table 1 shows the structured questionnaire we developed according to an extensive literature review (Huang, 2018, 2019, 2020b; Damjanovic et al., 2015; Liao et al., 2015; Mitchell, 1996; Tjosvold, 1988; Xu et al., 2014). This questionnaire comprises a total of seven constructs, including perceived competition, perceived collaboration, perceived enjoyment, perceived usefulness, attitude toward using, behavioral intention, and learning performance. The participants expressed their opinions about the game by filling in this

Table 1 The final questionnaire

Construct	Item
Perceived competition	PM1 We compete with the other group when playing the game. PM2 We want to know which group scores higher when playing the game. PM3 We try to outperform the other group when playing the game.
Perceived collaboration	PL1 My teammate and I collaboratively solve the challenges when playing the game. PL2 My teammate and I discuss the solutions to the challenges with each other when playing the game. PL3 I help my teammate complete his/her tasks when playing the game.
Perceived enjoyment	PE1 I find the game interesting. PE2 I find the game exciting. PE3 I find the game enjoyable.
Perceived usefulness	PU1 I learn about database conveniently by playing the game. PU2 I learn about database efficiently by playing the game. PU3 I can improve my performance if I learn about database by playing the game.
Attitude toward using	AT1 I find it fun to learn about database through the game. AT2 I think it's a good idea to learn about database through the game. AT3 It's more interesting to learn about database through the game.
Behavioral intention	BI1 I want to play the game frequently. BI2 I will recommend my classmates to play the game. BI3 I wish to play the game if possible.
Learning performance	LP1 I can increase my understanding about database by playing the game. LP2 I can raise my grades in the course on database by playing the game. LP3 I can improve my learning performance on database by playing the game.

questionnaire, in which they indicated their level of agreement with the statements using a five-point Likert scale.

Participants and data collection

The participants recruited to this study were 108 sophomores from 3 different classes in the Department of Information Management at a University of Science and Technology in Taiwan. This study employed convenient sampling to recruit the participants. That is, the teachers explained the purpose of this study to the students in the 3 classes and invited their participation. The data collection for this study proceeded in two parts. The first part was a 15-week course on database principles & design, through which the teacher imparted database-related knowledge to the participants. The contents of the course encompassed the entity-relationship model, normalization, and structured query language. The second part was a 2-week practice. The participants prepared for the final exam by using the IC2 game to conduct question bank practice. In this part, the teachers introduced the IC2 game to the participants and encouraged them to use it voluntarily in their preparation for the final exam. That is, in this study, the participants were still engaged in a conventional, teacher-centered pattern of learning, but they could use the IC2 game to review and examine their learning outcomes. More specifically, the IC2 game was a question bank practice game. It offered the participants multi-choice tests on database principles and design. The participants could use these tests to check whether they have understood database principles and design. Afterwards, the participants were required to complete the questionnaire, based on which their opinions about the IC2 game were collected and the influences of competition and collaboration on learning performance were analyzed.

Results

This study adopted the partial least squares structural equation modelling (PLS-SEM) to analyze the data collected from the questionnaire. Specifically speaking, the PLS-SEM is a method for causal relationship analysis. It assists researchers in understanding the causal relationships among different variables, and then helps them make corresponding decisions or suggestions. Meanwhile, the PLS-SEM can process samples of small size or non-normal distribution (Chin & Newsted, 1999), which is why it has been noticed by researchers and widely employed in various disciplines such as education, marketing, management, and social science (Ghasemy et al., 2020). More importantly, the PLS-SEM is well suited for theory building-oriented studies, such as adding new variables and investigating their causal relationships with existing ones (Wang et al., 2022). As a result, this study adopted the PLS-SEM to explore the correlations among competition, collaboration, and learning performance. Besides, this study used the SmartPLS 3 software to fulfill the PLS-SEM, which contained a measurement model and a structural model. The measurement model was intended to assess the reliability and validity of the questionnaire, while the structural model was aimed at evaluating the significance of each hypothesis formulated in this study.

Measurement model

The measurement model primarily provides the indicators for the reliability and validity of the questionnaire. The indicators for reliability include Cronbach's alpha and composite

reliability. The value of both indicators must exceed 0.70 for a construct to be reliable (Hair et al., 2006). The indicators for validity include convergent validity and discriminant validity. The former consists of item loadings and average variance extracted (AVE), while the latter the square root of AVE and Heterotrait-monotrait (HTMT) ratio. In terms of convergent validity, the minimum value of item loadings for a statement to be valid is 0.70 (Chin & Newsted, 1999); while the minimum value of AVE for a construct to be valid is 0.5 (Hair et al., 2006). In terms of discriminant validity, a construct is valid when its square root of AVE is larger than the correlation coefficients between it and other constructs (Fornell & Larcker, 1981); and the constructs are valid when the HTMT ratio of the correlations between the constructs is smaller than 0.90 (Henseler et al., 2015). Tables 2 and 3, and 4 demonstrate that the questionnaire were reliable and valid, since it met the criteria for reliability, convergent validity, and discriminant validity.

Structural model

Apart from using path coefficients and R^2 value to test the structural model, this study also employed standardized root mean square residual (SRMR) and normed fit index (NFI) to assess the goodness of model fit before testing the structural model. The model has high goodness of fit if its SRMR value is less than 0.10 or 0.08. The range of NFI value is between 0 and 1. The closer the NFI value is to 1, the higher goodness of fit the model has (Ringle et al., 2022). Accordingly, the goodness of fit of the model in this study was acceptable, because its SRMR and NFI values were 0.054 and 0.82 respectively. Path

Table 2 The reliability and convergent validity

Construct	Reliability		Convergent validity		
	Cronbach's alpha	Composite reliability	Items loading	AVE	
Perceived competition	0.81	0.89	PM1	0.87	0.72
			PM2	0.81	
			PM3	0.87	
Perceived collaboration	0.92	0.95	PL1	0.92	0.86
			PL2	0.94	
			PL3	0.93	
Perceived enjoyment	0.92	0.95	PE1	0.91	0.86
			PE2	0.93	
			PE3	0.95	
Perceived usefulness	0.95	0.97	PU1	0.95	0.91
			PU2	0.96	
			PU3	0.95	
Attitude toward using	0.94	0.96	AT1	0.95	0.89
			AT2	0.95	
			AT3	0.94	
Behavioral intention	0.93	0.96	BI1	0.93	0.88
			BI2	0.93	
			BI3	0.95	
Learning performance	0.96	0.97	LP1	0.96	0.93
			LP2	0.97	
			LP3	0.96	

Table 3. The discriminant validity by using the square root of AVE

Construct		1	2	3	4	5	6	7
Perceived competition	1	0.85						
Perceived collaboration	2	0.44	0.93					
Perceived enjoyment	3	0.55	0.49	0.93				
Perceived usefulness	4	0.44	0.63	0.61	0.95			
Attitude toward using	5	0.38	0.54	0.68	0.77	0.95		
Behavioral intention	6	0.41	0.45	0.75	0.61	0.78	0.94	
Learning performance	7	0.30	0.55	0.64	0.81	0.85	0.80	0.96

Table 4. The discriminant validity by using HTMT ratio

Construct		1	2	3	4	5	6	7
Perceived competition	1	–						
Perceived collaboration	2	0.51	–					
Perceived enjoyment	3	0.62	0.54	–				
Perceived usefulness	4	0.49	0.67	0.65	–			
Attitude toward using	5	0.42	0.58	0.73	0.81	–		
Behavioral intention	6	0.46	0.49	0.82	0.65	0.83	–	
Learning performance	7	0.33	0.59	0.68	0.84	0.89	0.84	–

coefficients revealed the tenability of the hypotheses, whilst R^2 value indicated the model's explanatory power over the variation in the dependent variables (Chin & Newsted, 1999). Figure 1 illustrates the results of the structural model: one hypothesis is rejected (i.e., H7), and the other eight hypotheses are confirmed. Figure 1 also shows the percentages of the variation in each of these constructs that the structural model can explain.

The results of the structural model are extended in Table 5 that exhibits the direct, indirect, and total effects of each construct, from which the decisive factors behind learning performance can be identified. In addition, Table 5 reveals two intriguing phenomena. The first phenomenon is that perceived competition, rather than perceived collaboration, exerts a more significant influence on learning performance. The second phenomenon is that perceived enjoyment has a stronger effect than perceived usefulness does on behavioral intention.

Discussions

Table 5 revealed the first finding of this study, i.e., perceived competition plays a more powerful role than perceived collaboration in affecting learning performance. Table 5 also showed that learning performance is directly affected by behavioral intention, and the latter is influenced by perceived enjoyment significantly and by perceived usefulness moderately. In addition, Fig. 4 proved that perceived enjoyment and perceived usefulness are respectively affected by perceived competition and perceived collaboration in

Table 5. The direct, indirect and total effects

Dependent variable	Independent variables	Direct effects	Indirect effects	Total effects
Learning performance	Perceived competition	–	0.34	0.34
	Perceived Collaboration	–	0.15	0.15
	Perceived enjoyment	–	0.48	0.48
	Perceived usefulness	–	0.24	0.24
	Attitude toward using	–	0.43	0.43
	Behavioral intention	0.80	–	0.80
Behavioral intention	Perceived competition		0.42	0.42
	Perceived Collaboration		0.19	0.19
	Perceived enjoyment	0.43	0.18	0.61
	Perceived usefulness	–	0.30	0.30
	Attitude toward using	0.54	–	0.54

a direct manner. From these results, we can infer that competition is more capable than collaboration in improving learning performance within IC2 contexts. This is because competition brings students enjoyment and collaboration makes them aware of the usefulness of DGBL. Nevertheless, enjoyment is a more crucial factor behind students' intention to engage in DGBL. In other words, students will engage in learning through the game because competition brings them pleasure, and thereby improves their learning performance. While this finding shows that competition has a more significant influence on learning performance compared to collaboration, it is important to note that this does not necessarily hold true in all DGBL. In some collaborative-based DGBL (Chang & Hwang, 2019), such as those that emphasize cooperative knowledge construction (Lee et al., 2016), competition is not a key factor in affecting learning performance. Therefore, it is possible that this finding is specific to IC2 contexts where competition plays a more critical role in influencing learning performance. Take this study for example. As the participants were taking the question bank practice through the IC2 game, they found the practice enjoyable because of its competition mechanism, and they were ergo willing to immerse in such practice, hence improvement of their learning performance. This inference echoes the behavioral decision theory. For instance, O'Curry and Strahilevitz (2001) pointed out that consumers tend to adopt emotional views about a product when they are not sure whether they should use it or not (for example, whether the product can bring them pleasure or not). In contrast, they tend to take rational views about a product when they determine to use it (for example, the product will improve their work performance). In this study, the participants were informed of the purpose of the IC2 game and invited to the experiment. At that time, they were not sure whether they should use it for learning in the future, and held an emotional view about the game (i.e. whether the game will bring them pleasure). Therefore, the participants adopted an ambivalent attitude toward the game. As a matter of fact, educational games are less mature than other types of educational technology. They have not been extensively applied to the real teaching scene yet. Students are usually uncertain about educational games and seldom adopt rational views about them. As a result, perceived enjoyment plays a more significant role than perceived usefulness in affecting students' attitude toward educational games (Huang, 2018, 2020b). In sum, perceived competition provides students with enjoyment, and perceived collaboration makes them aware of the

usefulness of DGBL, in which enjoyment exerts a more significant influence on learning performance through the mediation of behavioral intention.

Another interesting finding revealed in Fig. 4 was that perceived competition directly affects perceived collaboration. This finding suggests that the competition mechanism can stimulate intragroup collaboration in IC2 contexts. This is a very intriguing finding because it echoes Chen and Chiu (2016) and contrasts with ter Vrugte et al. (2015). First of all, ter Vrugte et al. (2015) found that competition may be detrimental to the collaboration among students who deliver poor or mediocre performance. Specifically speaking, ter Vrugte et al. (2015) originally expected competition to enhance the effectiveness of collaboration in DGBL contexts, because they believed that competition will increase intragroup interdependence and individual accountability. However, the results confounded their expectation. Students who deliver poor or mediocre performance need to invest most of their energy in understanding the competition mechanism when it is overly complicated, which prevents them from engaging in collaboration. Accordingly, they suggested reducing the complexity of competition mechanism in order to immerse students in collaboration without being disturbed by competition. On the other hand, Chen and Chiu (2016) proved that competition indeed facilitates students' collaboration, because a competition mechanism, such as team scores, prompts students to discern the intergroup differences and decide whether they should work harder for better performance. Meanwhile, they found that the existence of a competition mechanism will urge students to pursue a common objective, hence their interdependence and solidarity. Therefore, our finding echoes not only Chen and Chiu (2016) but also the social interdependence theory (Lewin, 1935). The theory claims that the essence of a group is the interdependence among its members who share a common goal, which makes the group a united whole. In this study, the mechanism of intergroup competition provided the participants in each group with a common goal, while the design of intragroup collaboration established the interdependence among teammates (i.e., the respondent was responsible for answering questions and the gamer for completing lines of tetrominoes, so that their group earned corresponding scores). To put it differently, the respondent and the gamer had their respective objectives, and their objectives were linked to the group's common goal, namely winning the game. As a result, the introduction

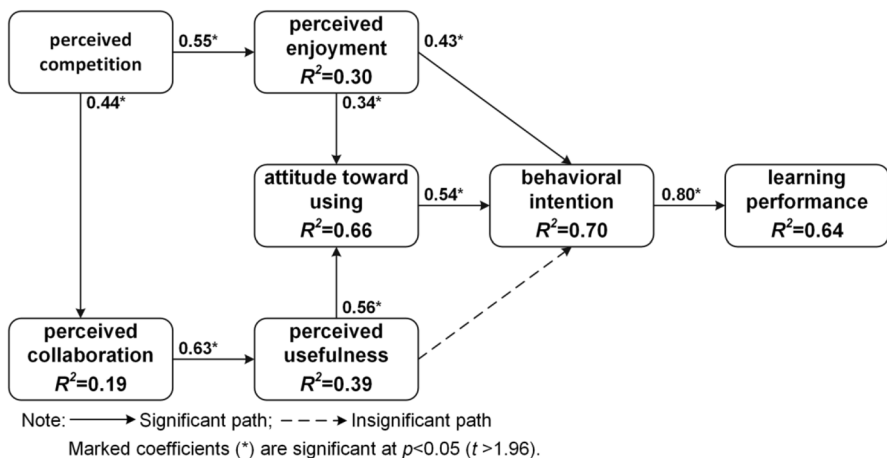


Fig. 4 The results of the structural model

of intergroup competition into the DGBL context fostered proactive and interdependent collaboration.

Figure 4 also revealed an unexpected finding, that is, in comparison with perceived enjoyment, perceived usefulness has no significant and direct influence but an indirect one on behavioral intention via attitude toward using. This finding implies that the usefulness of an IC2 game does not directly affect students' intention to play it, but exerts a significant and direct influence on their attitude that indirectly affects their intention to play it. This finding contradicts most studies on educational technology (Al-Rahmi et al., 2018; Garcia et al., 2019; Lin et al., 2021), but echoes studies on educational games (Huang, 2018, 2019, 2020b). In terms of educational technology, Al-Rahmi et al. (2018), Garcia et al. (2019) and Lin et al. (2021) have identified that perceived usefulness is not only the direct predictor of behavioral intention, but also the indirect one through the mediation of attitude toward using. In terms of educational game, Huang (2018; 2019) found that perceived usefulness only exerts indirect influence on behavioral intention via attitude toward using. Similarly, Huang (2020b) showed that the perceived enjoyment of educational games directly influences satisfaction in a way that perceived usefulness does not. A plausible explanation is that students may still use educational games for entertainment instead of learning. Educational games are not formal leaning tools. As a consequence, the usefulness of educational games has no direct influence on students' intention to use them. Nonetheless, comparing with other entertaining games, educational games afford students additional function for learning. Their usefulness ergo has a direct and significant influence on students' attitude toward them. According to the motivation theory, perceived usefulness belongs to extrinsic motivation, while perceived enjoyment an intrinsic one (Davis et al., 1992). In this study, the participants were invited to play the game for the purpose of experimentation. They were simply the participants in this study rather than learners who earnestly use the game for learning, hence their weaker extrinsic motivation. This inference also echoes the first finding of this study: the participants held an ambivalent attitude toward the game and therefore they adopted emotional views about it, which was possibly why perceived usefulness did not directly affect behavioral intention.

Conclusions

The role of competition in DGBL contexts has received considerable scholarly attention, whereas that of collaboration just came into researchers' spotlight for it can mitigate the negative effect produced by competition. However, a thorough study on how competition and collaboration influence learning performance in DGBL contexts remains wanting. To fill this gap in the literature, this study developed a research model to clarify their correlations. This model treated perceived competition and perceived collaboration as the independent variables, and learning performance the dependent one. Our research findings indicated that (1) both perceived competition and perceived collaboration exert significant influences on learning performance, among which perceived competition not only affects perceived collaboration directly, but are also more influential than perceived collaboration in affecting learning performance; (2) perceived competition has both direct and indirect influences on behavioral intention via perceived enjoyment; (3) perceived collaboration has an indirect influence on behavioral intention via perceived usefulness; and (4) behavioral intention exerts a significant and direct influence on learning performance. These outcomes imply that competition indeed brings students pleasure, while collaboration leads

to perceived usefulness. The entertainment provided by an educational game directly wins students' acceptance of the game, yet only perceived usefulness can indirectly influence their willingness to use the game. In comparison with collaboration, competition directly prompts students to use an educational game with the entertainment it offers, which is why competition is able to influence learning performance in a significant fashion. In sum, the results of this study not only help researchers grasp the correlations among competition, collaboration, and learning performance in DGBL contexts, but also facilitate the application and development of DGBL, thereby assisting students in producing better learning performance.

Two important lessons can be drawn from the research findings of this study. Firstly, it is quite crucial for game designers to devise simple and clear competition mechanisms, because such mechanisms not only galvanize students to engage in DGBL, but also facilitate their collaboration. In fact, competition in DGBL contexts is a double-edged sword. It can be both beneficial and counterproductive to students' learning performance. Previous studies have pointed out that students will be misguided by overly complicated competition mechanisms to concentrate on winning the game instead of learning from a game. Accordingly, game designers have to devise simple and clear competition mechanisms for students, so that they can devote most of their effort to learning. Besides, game designers need to link victory conditions with learning performance. In other words, students must know that learning is the only way to win the game. Moreover, game designers have to conceive a fair division of labor for the players in a team, so as to enable the team members to forge positive interdependence by competing with the other teams. The second lesson is that, for practitioners, guiding students to perceive the usefulness of an educational game will facilitate its promotion. Specifically speaking, students still regard educational games as the extension of games for entertainment, which is why the enjoyment brought by an educational game directly affects their willingness to use it, yet its usefulness does not. That is, the pleasure offered by an educational game is sufficient to induce students to use the game, but its perceived usefulness remains weaker in this regard. Therefore, practitioners should explicate the usefulness of an educational game in addition to the enjoyment it may bring, thereby allowing students to perceive the very purpose of the game; to wit, learning. Similarly, apart from working on the elements of entertainment, game designers should also try to promote the educational function of such games, thereby orientating students to adopt a correct attitude toward educational games. To put it another way, students should take educational games seriously as learning tools rather than merely innovative learning experiences.

Although this study advanced our understanding of the correlations among competition, collaboration, and learning performance in DGBL contexts, it still has some methodological limitations. Firstly, this study was conducted in Taiwan where students are placed under considerable pressure of credentialism. There is a popular stereotype in Taiwan that better educational backgrounds lead to brighter employment prospects (Chan & Lin, 2015; Chen et al., 2015). It implies that the participants in this study may have been accustomed to competition rather than collaboration. As a result, competition is the key driving force behind their intention of learning. In other words, our research findings may not be generalizable to other countries, particularly those don't attach special value to credentialism, such as the United States (Groß, 2003). Secondly, the game used in this study combines a question bank system with Tetris, which may limit the generalization of our research findings to this specific type of game. While the analysis results suggest that perceived competition is a more significant factor than perceived collaboration behind learning performance, this does not necessarily mean that competition is always stronger than

collaboration in determining learning performance in digital game-based learning. Different types of games, such as action games, role-playing games, or massively multiplayer online role-playing games, may generate varying forms of interaction among players, making collaboration more important than competition in some games. Therefore, our findings cannot be generalized to all game-based learning cases. To address this limitation, we plan to design other types of games in future research to examine the generalizability of our findings. Thirdly, this study did not consider students' game preference, which will affect their attitude toward a given game and may lead to divergent results. Therefore, we will include students' game preference in our future research as a moderator variable and examine its influence. Fourthly, this study was done using 108 students sampled from a single department. We will take the participants' age, gender, and department into consideration in our future research, so as to examine whether their backgrounds will affect their performance in DGBL contexts. Fifthly, this study adopted self-report to measure learning performance instead of using exam papers to measure actual learning outcomes. We will try to design an exam paper with moderate difficulty and high discrimination, so as to further explore the influence of competition and collaboration on learning performance in DGBL contexts. Meanwhile, we will redesign the game by incorporating the monitoring and recording functions capable of capturing students' gaming processes, which will give us a deeper understanding of the correlations among competition, collaboration, and learning performance. Finally, we will enlarge our sample size in order to yield a more convincing argument.

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Ding-Chau Wang **Ding-Chau Wang** received Ph.D. degree in computer science and information engineering from National Cheng Kung University, Tainan, Taiwan. He is currently a professor in the Department of Information Management at Southern Taiwan University of Science and Technology, Tainan, Taiwan. His research interests include educational technology and game.

Yong-Ming Huang **Yong-Ming Huang** is currently a Professor in the Department of Multimedia and Entertainment Science at Southern Taiwan University of Science and Technology, Taiwan. He received his Ph.D. degree in Engineering Science from the National Cheng Kung University, Taiwan in 2012. His research interests include educational technology and game.